

Assignment 1: BGP Routing with ns-3

Course: Advanced Computer Networks(COL724/7524)

Summary

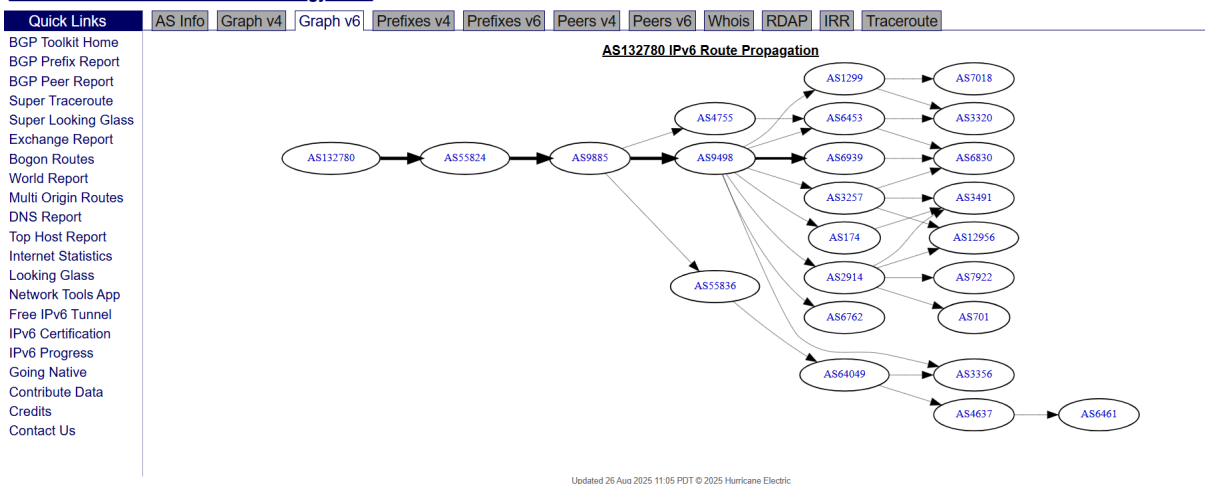
The prefix **103.27.9.0/24**, belonging to **AS132780 (IIT Delhi)**, is announced to the Internet through its upstream provider **AS55824 (NKN Core)**. This in turn connects to **AS9885 (NKN Gateway)**, which further connects with large Indian ISPs such as **AS4755 (Tata Communications)** & **AS9498 (Airtel)**, along with other NKN networks (**AS55836**).

These networks then interconnect with major global Tier-1 providers such as **AS6453 (Tata Global)**, **AS1299 (Telia)**, **AS3356 (Lumen/Level3)**, and **AS174 (Cogent)**.

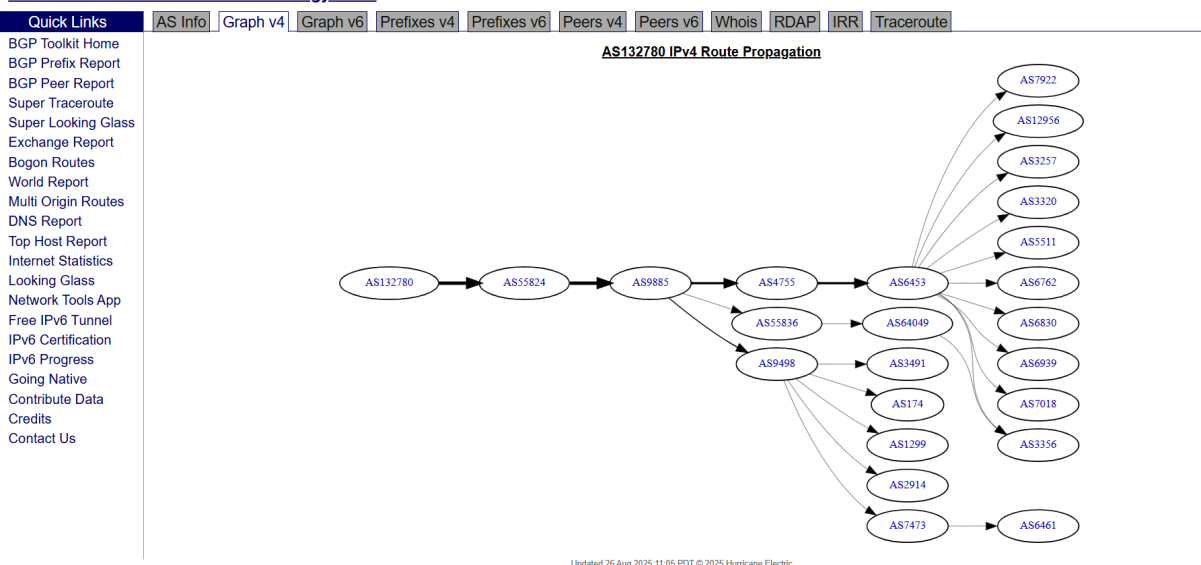
Thus, IIT Delhi's routes are globally reachable via a multilevel provider hierarchy, starting from NKN and extending into Tier-1 transit providers.

Some Screenshots of the connections on v4 & v6:

AS132780 Indian Institute of Technology Delhi



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Record:

Which upstream ISPs announce IIT Delhi's routes to the global Internet?

- **AS55824 (NKN Core)** is the direct upstream for **AS132780 (IIT Delhi)**.
- That in turn connects to **AS9885 (NKN Internet Gateway)**.
- From there, announcements go through **AS4755 (Tata Communications)**, **AS9498 (Airtel)**, and **AS55836 (NKN Access Network)**.

From which providers those ISPs themselves receive announcements (3 steps further)?

- AS4755 (Tata Comm) → upstreams include **AS6453 (Tata Global)**, which peers with Tier-1s like **AS7922 (Comcast)**, **AS12956 (Telefonica)**, **AS3257 (GTT)**, **AS3320 (Deutsche Telekom)**, **AS3356 (Lumen/Level3)**
- AS9498 (Airtel) → peers with **AS3491 (PCCW Global)**, **AS174 (Cogent)**, **AS1299 (Telia)**, **AS2914 (NTT)**, **AS3356 (Level3)**, etc.
- AS55836 (NKN Access) → propagates further through large ISPs (shown in graph: AS2914, AS7473, etc.).